

Fundamental Theorem of Arithmetic in $\mathbb{Q}[x]$

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1. Introduction

Theorem 1. *Theorems are massive results.*

You can also hyperlink to Theorem [1](#) if you want!

Corollary 2. *Corollaries follow from theorems.*

Remember to follow the good writing guidelines! You can also break pages if you feel it appropriate.

2. History

Historical background is often nice.

3. Axioms

You can make subsections if you find it appropriate.

3.1. Axioms of this type

These axioms explain these properties of the object. They are important because of these reasons.

3.2. Axioms of another type

These are different type of axioms from the previous one because of this. These are important because of these reasons.

4. Math

Definition 1. For $f(x) \in \mathbb{Q}[x]$, let the **norm**, or $\|f(x)\|$ be equal to

$$\|f(x)\| = 2^{\deg f(x)}$$

with the following properties:

- $\|f(x)\| \cdot \|g(x)\| = \|f(x)g(x)\|$

Proof.

- For any $c(x) \in \mathbb{Q}[x]^\times$, $\|c(x)\| = 1$.

Proof. Since $\deg c(x) = 0$, $\|c(x)\| = 2^0 = 1$.

5. Notation

Definition 2. Define **summation**, or $\sum_{i=i_0}^n a_i$ for $n \in \mathbb{Z}^+$ and $n \geq i_0$ as follows:

$$\sum_{i=i_0}^{i_0} = a_{i_0} \quad \sum_{i=i_0}^n a_i = a_n + \sum_{i=i_0}^{n-1} a_i$$

Lemma 3. $\sum_{i=i_0}^n a_i + \sum_{i=i_0}^n b_i = \sum_{i=i_0}^n (a_i + b_i)$

Proof. Note that

$$\sum_{i=i_0}^{i_0} a_i + \sum_{i=i_0}^{i_0} b_i = a_{i_0} + b_{i_0} = \sum_{i=i_0}^{i_0} (a_i + b_i)$$

Let $S = \left\{ k \in \mathbb{Z}^+ \mid \sum_{i=i_0}^{i_0+k} a_i + \sum_{i=i_0}^{i_0+k} b_i \neq \sum_{i=i_0}^{i_0+k} (a_i + b_i) \right\}$. For the sake of contradiction, S is not empty.

As $S \subseteq \mathbb{Z}^+$, by the Well-Ordering Principle, there exists a least element $l \in S$.

If $k = 1$:

$$\sum_{i=i_0}^{i_0+1} a_i + \sum_{i=i_0}^{i_0+1} b_i = a_{i_0+1} + a_{i_0} + b_{i_0+1} + b_{i_0} = (a_{i_0+1} + b_{i_0+1}) + (a_{i_0} + b_{i_0}) = \sum_{i=i_0}^{i_0+1} (a_i + b_i)$$

Thus, $1 \notin S$ and so $l > 1$ by NIBZO. Since $l - 1 \in \mathbb{Z}$ yet $l - 1 \notin S$,

$$\begin{aligned} \sum_{i=i_0}^{i_0+l-1} a_i + \sum_{i=i_0}^{i_0+l-1} b_i &= \sum_{i=i_0}^{i_0+l-1} (a_i + b_i) \\ a_l + b_l + \sum_{i=i_0}^{i_0+l-1} a_i + \sum_{i=i_0}^{i_0+l-1} b_i &= a_l + b_l + \sum_{i=i_0}^{i_0+l-1} (a_i + b_i) \\ a_l + \sum_{i=i_0}^{i_0+l-1} a_i + b_l + \sum_{i=i_0}^{i_0+l-1} b_i &= (a_l + b_l) + \sum_{i=i_0}^{i_0+l-1} (a_i + b_i) \\ \sum_{i=i_0}^{i_0+l} a_i + \sum_{i=i_0}^{i_0+l} b_i &= \sum_{i=i_0}^{i_0+l} (a_i + b_i) \end{aligned}$$

However, since $l \in S$, this is a contradiction. Thus, S must be empty. $\square$

Lemma 4. $\sum_{i=i_0}^n ca_i = c \sum_{i=i_0}^n a_i$ *Proof.* Note that

$$\sum_{i=i_0}^{i_0} ca_i = ca_{i_0} = c \sum_{i=i_0}^{i_0} a_i$$

and that

$$\sum_{i=i_0}^{i_0+1} ca_i = ca_{i_0} + ca_{i_0+1} = c \sum_{i=i_0}^{i_0+1} a_i$$

Let $S = \left\{ k \in \mathbb{Z}^+ \mid \sum_{i=i_0}^{i_0+k} ca_i \neq c \sum_{i=i_0}^{i_0+k} a_i \right\}$. For the sake of contradiction, S is not empty. As $S \subseteq \mathbb{Z}^+$,

by the well-ordering principle, there exists a least element $l \in S$. Since $1 \notin S$, $l > 1$ by NIBZO. As $l - 1 \in \mathbb{Z}$ yet $l - 1 \notin S$,

$$\begin{aligned} \sum_{i=i_0}^{i_0+l-1} ca_i &= c \sum_{i=i_0}^{i_0+l-1} a_i \\ ca_l + \sum_{i=i_0}^{i_0+l-1} ca_i &= ca_l + c \sum_{i=i_0}^{i_0+l-1} a_i \\ \sum_{i=i_0}^{i_0+l} ca_i &= c \sum_{i=i_0}^{i_0+l} a_i \end{aligned}$$

However, since $l \in S$, this is a contradiction. Thus, S must be empty. $\square$

Definition 3. Define **product**, or $\prod_{i=i_0}^n a_i$ for $n \in \mathbb{Z}^+$ and $n \geq i_0$ as follows:

$$\prod_{i=i_0}^{i_0} = a_{i_0} \quad \prod_{i=i_0}^n a_i = a_n \cdot \prod_{i=i_0}^{n-1} a_i$$

Lemma 5. $\prod_{i=i_0}^n a_i \prod_{i=i_0}^n b_i = \prod_{i=i_0}^n a_i b_i$

Proof. Note that

$$\prod_{i=i_0}^{i_0} a_i \prod_{i=i_0}^{i_0} b_i = a_{i_0} b_{i_0} = \prod_{i=i_0}^{i_0} a_i b_i$$

Let $S = \left\{ k \in \mathbb{Z}^+ \mid \prod_{i=i_0}^{i_0+k} a_i \prod_{i=i_0}^{i_0+k} b_i \neq \prod_{i=i_0}^{i_0+k} a_i b_i \right\}$. For the sake of contradiction, S is not empty. As $S \subseteq \mathbb{Z}^+$, by the Well-Ordering Principle, there exists a least element $l \in S$.

If $k = 1$:

$$\prod_{i=i_0}^{i_0+1} a_i \prod_{i=i_0}^{i_0+1} b_i = a_{i_0+1} a_{i_0} \cdot b_{i_0+1} b_{i_0} = (a_{i_0+1} b_{i_0+1})(a_{i_0} b_{i_0}) = \prod_{i=i_0}^{i_0+1} a_i b_i$$

Thus, $1 \notin S$ and so $l > 1$ by NIBZO. Since $l - 1 \in \mathbb{Z}$ yet $l - 1 \notin S$,

$$\begin{aligned}
& \prod_{i=i_0}^{i_0+l-1} a_i \prod_{i=i_0}^{i_0+l-1} b_i = \prod_{i=i_0}^{i_0+l-1} a_i b_i \\
& a_l b_l \cdot \prod_{i=i_0}^{i_0+l-1} a_i \prod_{i=i_0}^{i_0+l-1} b_i = a_l b_l \cdot \prod_{i=i_0}^{i_0+l-1} a_i b_i \\
& (a_l \prod_{i=i_0}^{i_0+l-1} a_i) \cdot (b_l \prod_{i=i_0}^{i_0+l-1} b_i) = (a_l b_l) \prod_{i=i_0}^{i_0+l-1} a_i b_i \\
& \prod_{i=i_0}^{i_0+l} a_i \prod_{i=i_0}^{i_0+l} b_i = \prod_{i=i_0}^{i_0+l} a_i b_i
\end{aligned}$$

However, since $l \in S$, this is a contradiction. Thus, S must be empty. $\square$

6. Conclusion

Conclude with the big ideas!